

IBM Data Analyst Capstone Project

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11/03/2026



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EXECUTIVE SUMMARY



- To remain competent in the global IT industry, continuous development and upskilling is essential
- Data analysis was performed to identify key trends and skills needed in the IT sector
- Data was extracted from the Stack Overflow Developer Survey, Github and IBM website, and subsequently cleaned and used for exploratory data analysis and creating visualizations
- Key findings show that HTML/CSS, JavaScript and TypeScript were the top programming languages for the current year and PostgreSQL was holding its ground as a preferred database in the current and upcoming years
- 35–44-year-olds were the main respondents of the survey, and most were from the North-American region



INTRODUCTION



- This report is intended at consolidating visualizations, dashboards and findings in relation to the Stack Overflow Developer Survey data
- The following was investigated:
 - Most in-demand programming languages
 - Most in-demand databases
 - Most popular webframes
- The report is targeted at any business or IT professionals that may need to make informed decisions based on in-demand IT skills and the related demographics



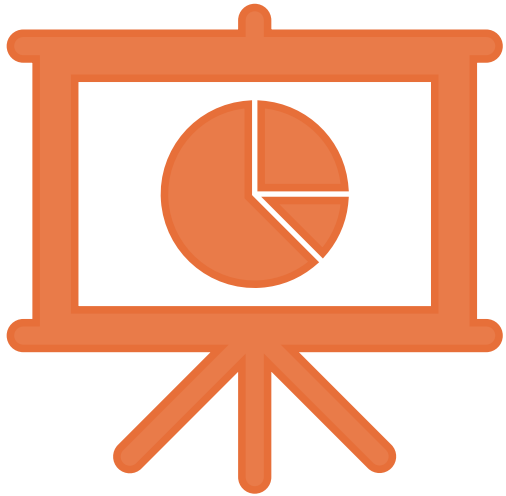
METHODOLOGY



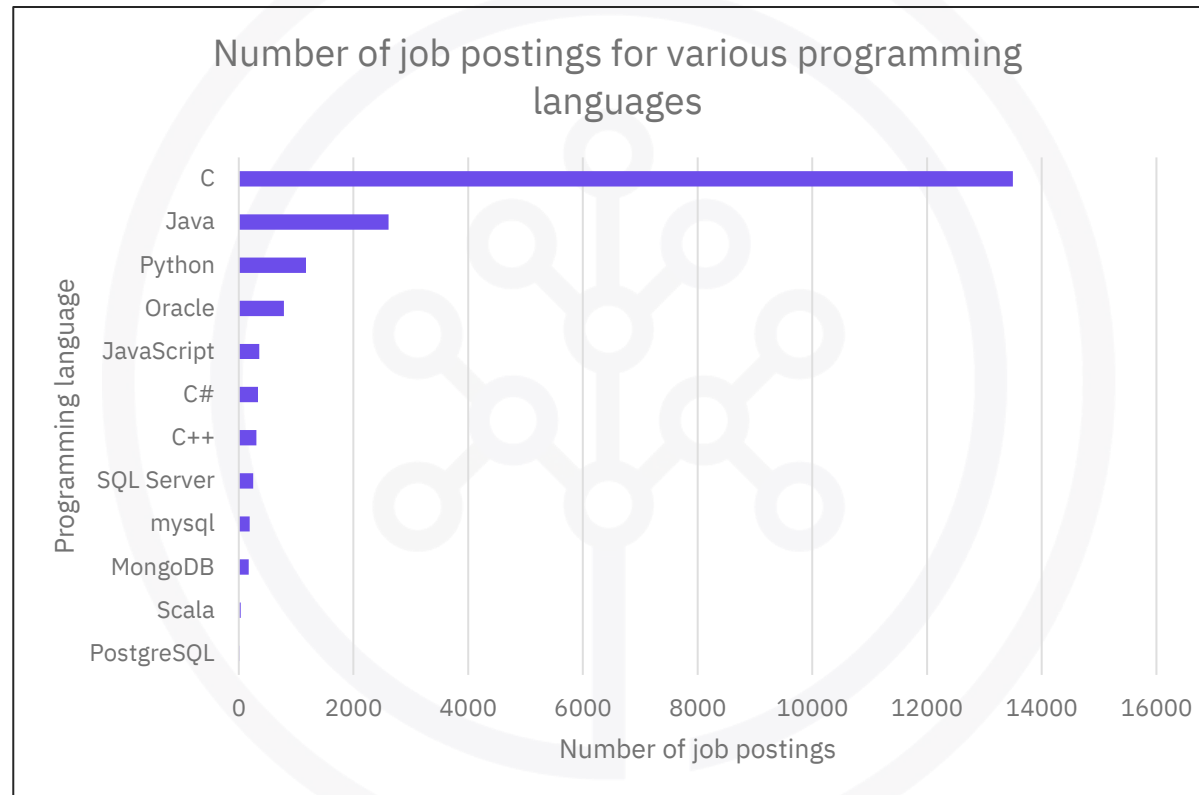
- Github Jobs API was used to source data on the number of job postings available for various programming technologies
- An average annual salary was extracted on the IBM website for the various technologies by utilizing web scraping techniques
- Python was predominantly used for data cleaning and analysis
- Python and IBM Cognos Analytics were used to generate visualizations and build dashboards



Results: Visualization – Charts



JOB POSTINGS

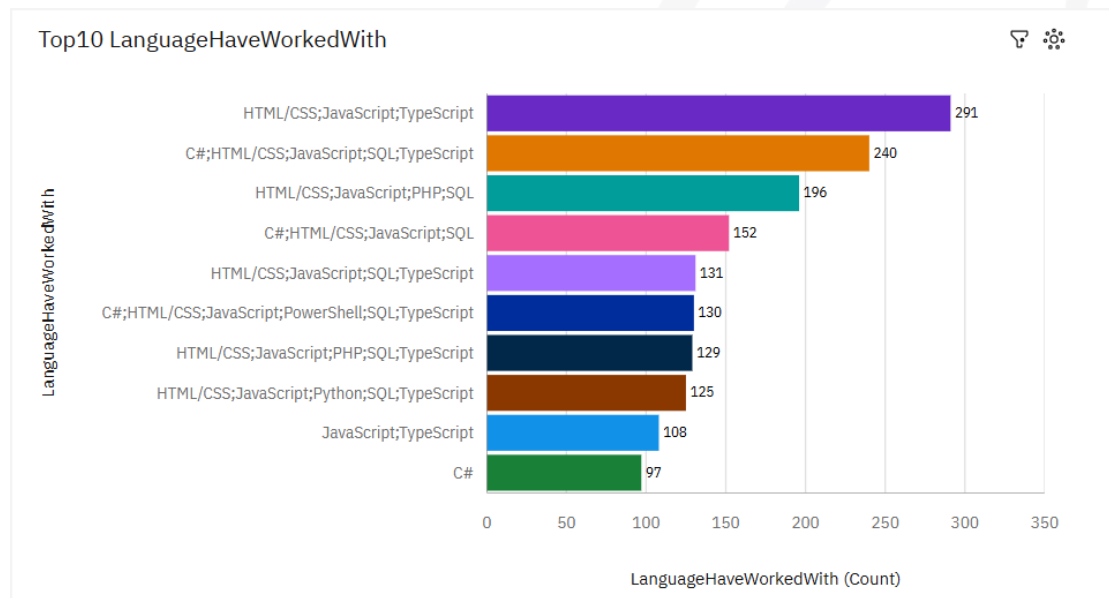


POPULAR LANGUAGES

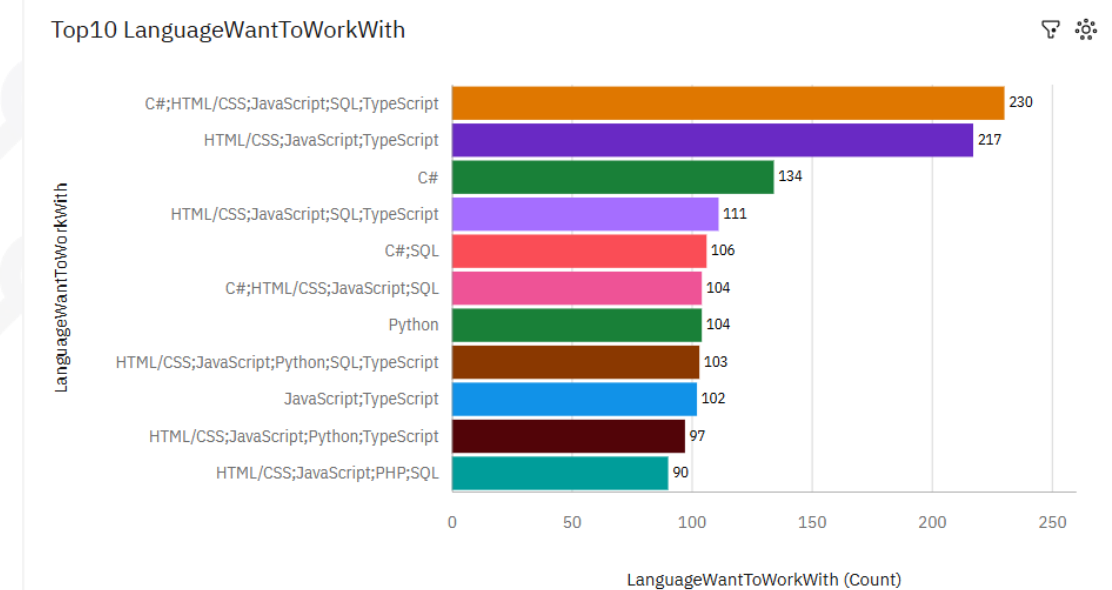


PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- HTML/CSS, JavaScript and TypeScript the most utilized languages in the current year
- C#, HTML/CSS, JavaScript, SQL and TypeScript to become the most used languages in subsequent years
- More languages will be in demand in the coming years in comparison to current year

Implications

- There is a high demand for web development skills as HTML and JavaScript are commonly used for this purpose
- SQL to become a required language due to the increasing demand for querying skills, and C# for its diversity across various applications
- The playing ground will be more competitive as more languages are adopted over the years

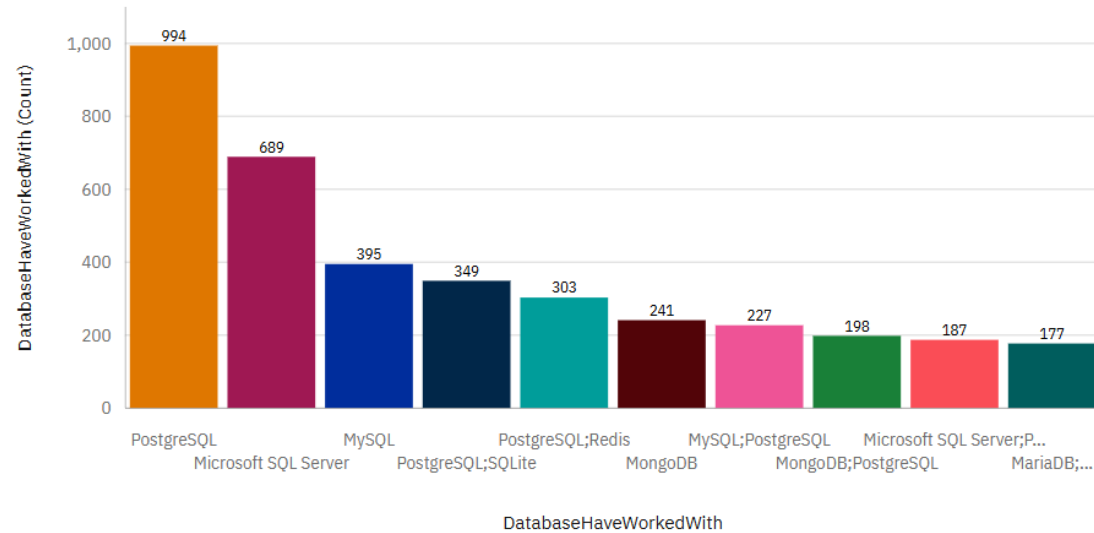


DATABASE TRENDS

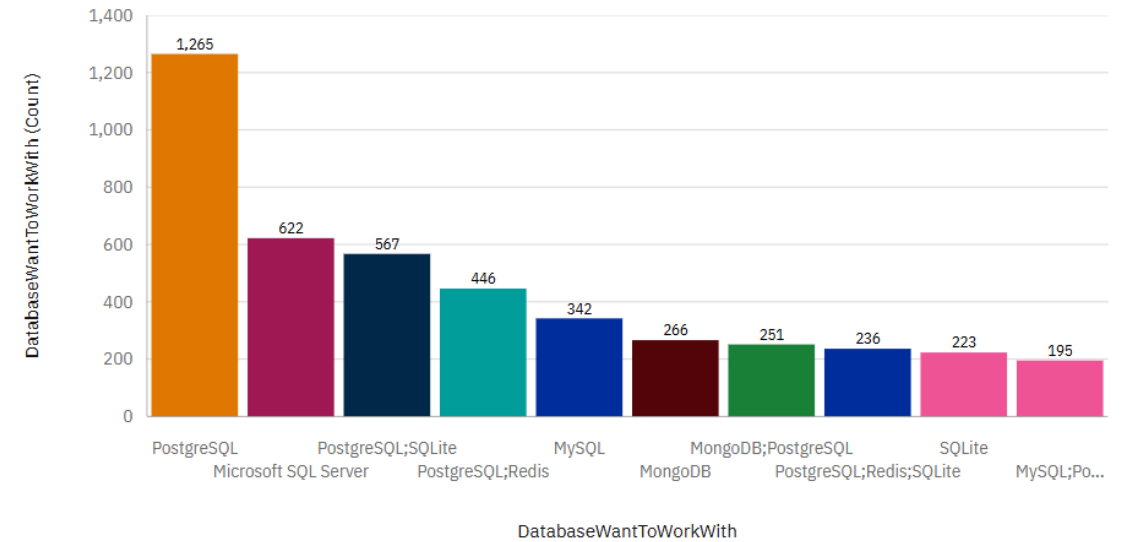
Current Year

Next Year

Top10 DatabaseHaveWorkedWith



Top10 DatabaseWantToWorkWith



DATABASE TRENDS - FINDINGS & IMPLICATIONS

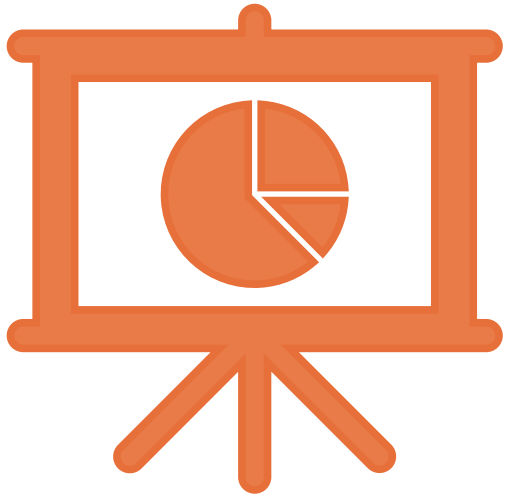
Findings

- PostgreSQL, Microsoft SQL Server and MySQL are the most used databases in the current year
- PostgreSQL and SQLite to replace MySQL as the third most used database in the coming years
- PostgreSQL to increase in demand in subsequent years

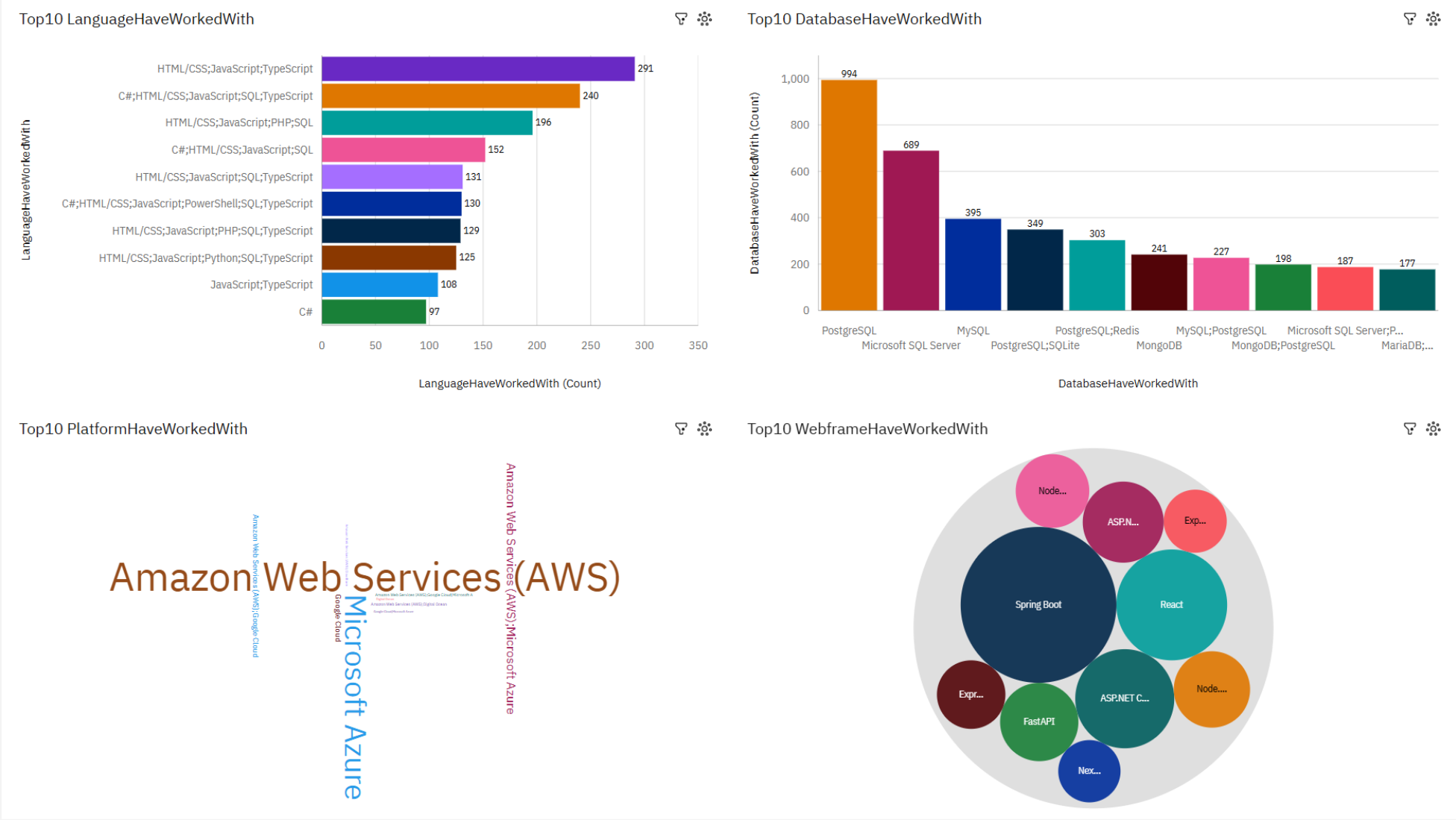
Implications

- SQL is still very much in demand in the current and coming years
- PostgreSQL and SQLite gaining more popularity over MySQL due to their scalability and security, making them suitable where data integrity is critical
- PostgreSQL seems to increase in popularity due to its open-source nature, robust feature set and high reliability

Results: Dashboards

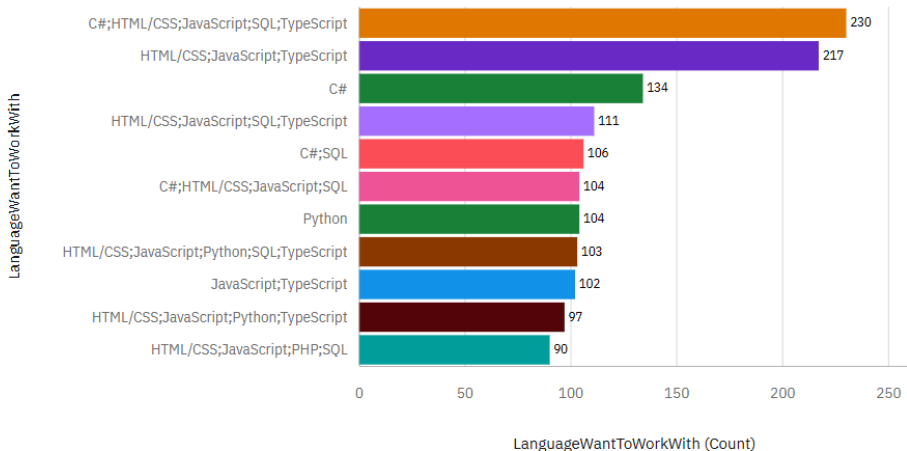


DASHBOARD TAB 1: Current Technology Usage

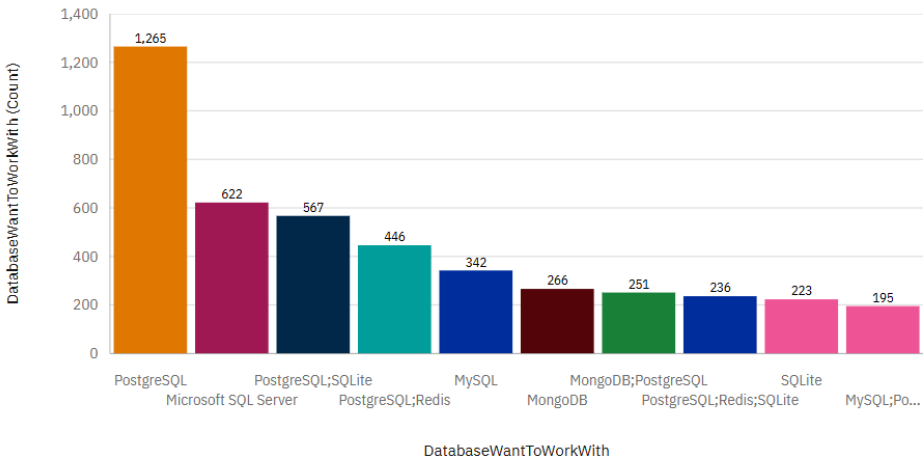


DASHBOARD TAB 2: Future Technology Trends

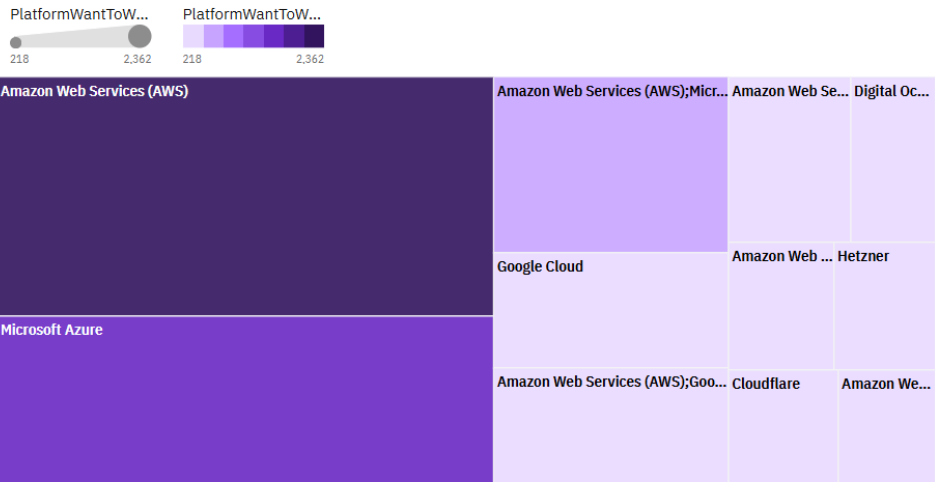
Top10 LanguageWantToWorkWith



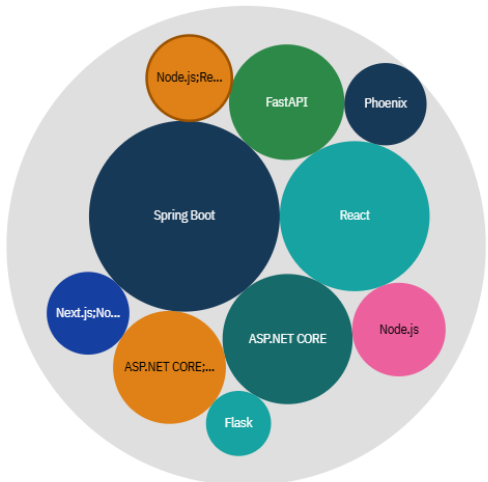
Top10 DatabaseWantToWorkWith



Top 10 PlatformWantToWorkWith



Top 10 WebframeWantToWorkWith

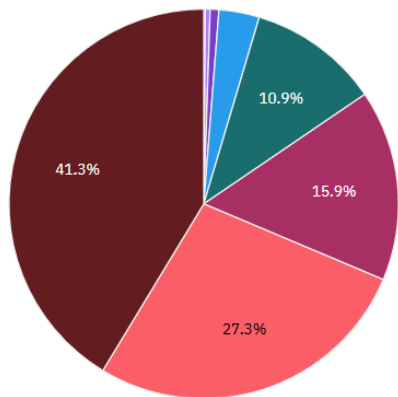


DASHBOARD TAB 3: Demographics

Respondent distribution by Age

Age

- Prefer not to say
- 65 years or older
- Under 18 years old
- 55-64 years old
- 45-54 years old
- 18-24 years old
- 35-44 years old
- 25-34 years old

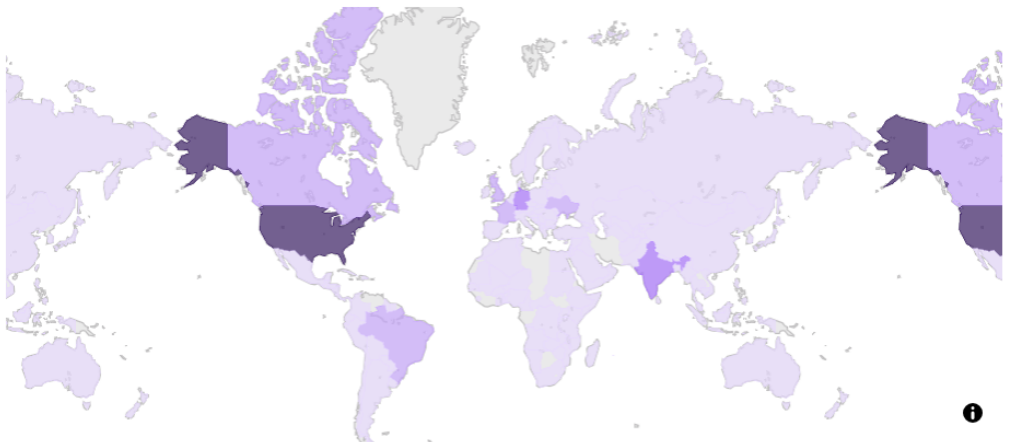


Respondent Count by Country

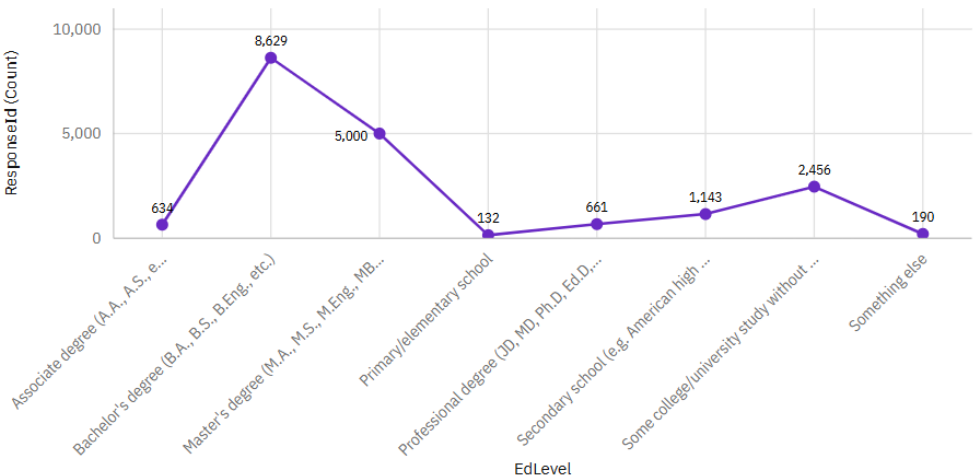


ResponseId (Count)

1 3,441



Respondent distribution by Formal Education Level

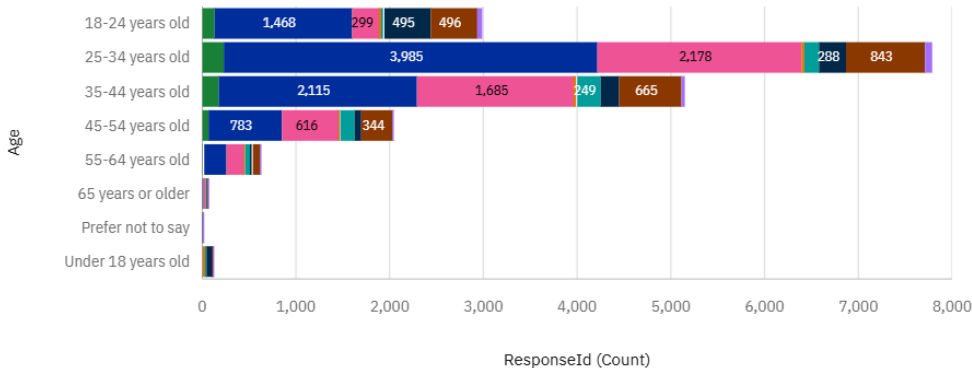


Respondent Count by Age, classified by Education Level



EdLevel

- Associate degree (A.A., A.S., etc.)
- Primary/elementary school
- Some college/university study without earning a degree
- Bachelor's degree (B.A., B.S., B.E., etc.)
- Professional degree (JD, MD, Ph.D., etc.)
- Something else
- Master's degree (M.A., M.S., M.En., MB.A., etc.)
- Secondary school (e.g. American high school)



Skill



DISCUSSION



- Upskilling and continuous development is critical to stay relevant in the growing IT space
- Amazon Web Services (AWS) remains a preferred cloud computing service
- Out of those who disclosed their ages, the 35–44-year-old range is prominent amongst the respondents and in the IT sector
- Africa, Asia and Australasia have the least respondents which indicates a room for growth and development in those regions



OVERALL FINDINGS & IMPLICATIONS

Findings

- Most respondents in the study have a bachelor's degree at the least
- Most professionals in the space fall within the 35–44-year-old age category
- Languages involving web development seem to be the most in-demand

Implications

- You are most likely to make it in the industry if you have a bachelor's degree at the least
- Continuous development is imperative to stay relevant and competent in a highly competitive IT sector
- More development is needed in under-developed regions like Africa, Asia and Australasia

CONCLUSION



- Technological advancement requires one to upskill and continuously develop their skills in an everchanging tech environment